Introduction to programming

CSE 101 Section 03

Introduction to Algorithms and Computing

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What is programming?

- A simple answer would be, "Programming is the act of instructing computers to carry out tasks." It is often referred to as **coding**.

- So then, what is a **computer program**?

- A computer program is a sequence of instructions that the computer executes.

- A computer is any device that is capable of processing code. This includes smartphones, ATMs, servers to name a few.

The Natural Language of the Computer

- Machines have their natural language like humans do.
- Computers *do not understand* the human language.

- The natural language of computers is the binary code — 1 and 0. These represent two states: **on (1)** and **off (0)**.
  - That is the natural language of electronic equipment. It would be hectic for us as humans to communicate with the computer in binary.

Enter Programming Languages

- To communicate with machines who speak binary, we do so in a language that's closer to our own natural language, such as English or French.

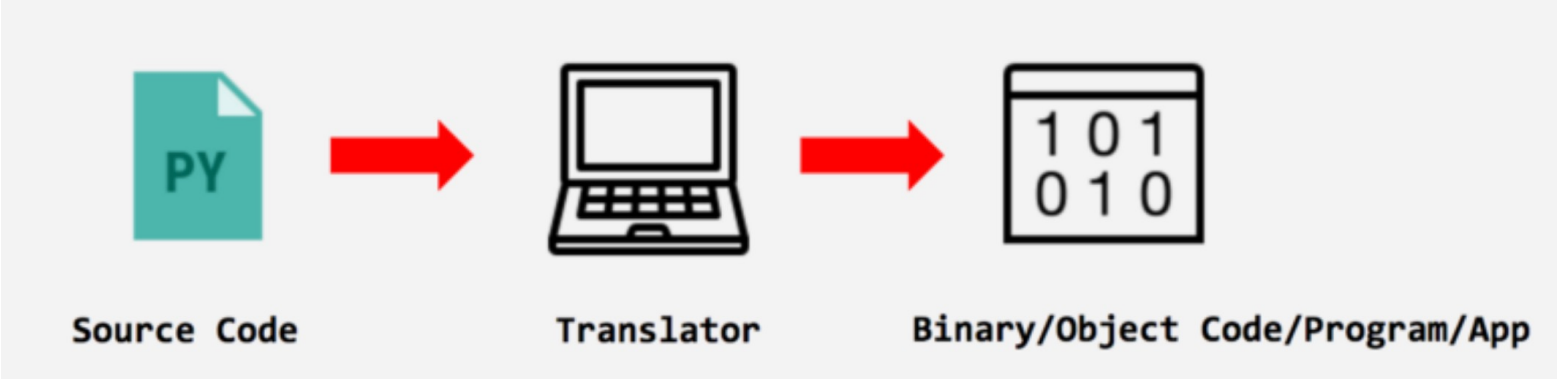
- Programming languages are *close* to our natural languages. But they are more structured and must be thoroughly learned.

- The computer needs a way to understand our human language. To do this, we'll need a translator!

What are Translators?

- Source code refers to code written in a particular programming language.

- Translators have the responsibility of converting your source code to the machine language. This is also known as **binary**. Remember ones and zeros. We may refer to the binaries as **Object Code** , the Program or a common word today: **App**.



More on Translators

Translators can be any of:

- Interpreters: a program that translates and executes high-level source code line by line, and immediately executes the translated lines.
- Compilers: a program that translates high-level source code into machine code, converting the entire code at once before execution.
- A hybrid of Interpreters and Compilers

Back to source code

- Now that you understand the concept of programming, we will examine source code.

- A source code is basically a file, just like a Microsoft (.docx) file, but slightly different.

- Depending on the language you are writing, there are designated extensions for saving your source code file. **Python**'s extension is '**.py**'. Java is '.java'. Matlab is '.m' to name a few.

Python

- In this course, we will learn programming using the language Python.

- Python was developed as a hobby project by Guido Van Rossum in December 1989, while he was looking for a hobby project to keep him occupied in the week around Christmas.

- Python is not named after the snake, but rather the British comedy group Monty Python.

Python

- Readable and simple compared to other programming languages
- Whitespace (indentation) matters and has a meaning.
- Cross platform: can be used on various operating systems (e.g. Windows, macOS, and Linux)
- One of the most popular programming languages.

Python Environment Setup

- We will be using a software called PyCharm.
- Think of it as a text editor to write Python code, but it does more than a simple text editor.
- In today's lab, we will discuss more about installation and use of PyCharm.

- Make sure you have your lab computer accounts setup, or bring your own laptops if you have one.

Example

```
a = 2
b = 3

sum = a + b

print(f"the sum of {a} and {b} is {sum}")
```

Example - modified

```
a = int(input("Enter the first number: "))
b = int(input("Enter the second number: "))

sum = a + b

print(f"the sum of {a} and {b} is {sum}")
```

Some apps/software that use Python

Next…

Introduction to Algorithms